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Long-term Outcomes of Multiligament Knee Injuries in American Football Players

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Background: Long-term outcomes for isolated anterior cruciate ligament (ACL) reconstructions in competitive American football athletes are well reported in the literature, but little data currently exist regarding multiligament knee injury (MLKI) reconstruction outcomes.

Purpose: To examine patient-reported and return-to-sport outcomes of competitive American football athletes who underwent primary, single-staged, multiligament knee reconstruction.

Study Design: Case series; Level of evidence, 4

Methods: We identified patients from our institution's prospectively collected data repository between 2001 and 2020 who underwent single-staged surgical reconstruction of an MLKI sustained during competitive participation in American football. We assessed patient-reported outcomes at a minimum of 2 years after surgery using the International Knee Documentation Committee (IKDC) Subjective Knee Form and questions regarding surgical satisfaction and return to sport. Successful return to sport was defined as a return to preinjury level of competition. We summarized all outcome data and compared outcomes between 2-ligament and >2-ligament groups and between ACL-only MLKI injury and bicruciate MLKI injury groups using independent *t* test for IKDC scores and chi-square test for return to sport. Additionally, we evaluated predictors of postoperative IKDC scores using linear regression and predictors of return to sport using logistic regression.

Results: Outcome data were successfully collected for 53 of 73 total eligible patients (73%; mean follow-up time, 7.7 ± 4.0 years; all male; mean age at surgery, 18.1 ± 2.7 years). The mean postoperative IKDC score was 84 ± 16. The most common level of preinjury competition was high school (n = 36; 68%), followed by college (n = 10; 19%). Seven patients did not return to sport competition at any level due to limitations from their knee surgery, and 82% of patients that attempted to return to preinjury level of sport were able to do so. A total of 50 patients (94%) were satisfied or very satisfied with their surgical outcome. The 2-ligament (n = 39) and >2-ligament (n = 14) groups did not significantly differ in IKDC scores (P = .96) or proportions with successful return to sport (P = .77). Similarly, the ACL-MLKI injury (n = 39) and bicruciate MLKI injury (n = 14) groups did not significantly differ in IKDC scores (P = .89) or proportions with successful return to sport (P = .77). Age and body mass index were not significantly associated with IKDC scores or successful return to sport at follow-up (all $P > .05$).

Conclusion: This study may represent the largest cohort of competitive American football athletes evaluated for longitudinal outcomes after multiligament knee reconstruction. Despite the severity of these injuries, we found good knee-related function and that the large majority of athletes who attempted to return to sport were successful. The majority of athletes (94%) were satisfied with their operative treatment.

Keywords: football; multiligament knee injury; multiligament knee reconstruction; knee dislocation; case series; return to play

Studies examining mid- to long-term patient-reported outcomes and return to sport (RTS) after isolated anterior cruciate ligament (ACL) reconstructions (ACLRs) are well reported in the orthopaedic literature for competitive American football athletes, especially at the professional level.[8,24,25,34,36,40,41] In contrast, data regarding outcomes of multiligament knee reconstructions for multiligament knee injuries (MLKIs) in football athletes are lacking. This is likely because of the relative rarity of MLKIs when compared with isolated ACL injuries.[29] Often occurring via knee subluxation or frank dislocation, MLKIs are defined as any combination of injuries to the ACL, posterior cruciate ligament (PCL), medial collateral ligament (MCL), and lateral collateral ligament (LCL) and/or structures of the posterolateral corner (PLC), with a cruciate ligament almost always involved.[23,33] These injuries occur by low- or high-energy trauma, such as in sports or in motor vehicle accidents, respectively, and can prove devastating with a high risk of neurovascular complications, damage to

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the periarticular structures, and overall impaired knee joint mechanical function resulting from the injury.[3,10,17,22,37,38]

The few studies pertaining to mid- and/or long-term outcomes for surgically managed MLKIs that do exist have demonstrated increased complication and revision rates and worse patient-reported outcomes and RTS rates compared with primary, isolated ACLRs.[5,13] In the only publication to our knowledge to specifically examine RTS in competitive American football athletes after operative treatment of MLKIs, Bakshi and colleagues[5] reported that 64% of National Football League (NFL) athletes (n = 50) successfully returned to sports participation. Conversely, in the largest review of NFL athletes after isolated ACLR (n = 165), Mai and colleagues[31] reported that approximately 82% successfully returned to football over a 10-year period. In a recent systematic review, Dean and colleagues[10] evaluated outcomes across 15 studies following MLKIs sustained through a low-energy mechanism (eg, sports injury) and reported an International Knee Documentation Committee (IKDC) Subjective Knee Form score of 68.4; however, no studies to our knowledge have evaluated IKDC outcomes specific to American football athletes with operatively treated MLKIs.

Clearly, additional evidence is needed to better understand functional and RTS outcomes for competitive American football athletes, not only at the professional level but also at the collegiate and high school levels, who underwent operative treatment for an MLKI. Additionally, evaluating the effect that baseline patient characteristics, clinical and injury history, and injury severity have on outcomes may help us to better determine prognosis after surgery and clinical decision making for these athletes. In this study, our primary aim was to evaluate longitudinal patient-reported and RTS outcomes for competitive American football athletes of all levels who underwent multiligament knee reconstruction for an MLKI sustained via participation in American football over a 20-year period. Our secondary aims were to compare outcomes between athletes who underwent surgery for 2 ligaments versus those who had surgery for >2 injured ligaments, to compare outcomes between those with ACL-MLKI injuries (no PCL involvement) and bicruciate MLKI injuries, and to examine the associations between baseline patient and injury variables and outcomes.

METHODS

Participants

Institutional review board approval (SterlingIRB; IRB No. 9023) was obtained before the initiation of this study. This case series has been reported in line with the Preferred Reporting Of CasESeries in Surgery guidelines.[1] A prospectively collected database was used to retrospectively identify potential study participants based on diagnosis and procedures between January 2001 and August 2020 at Andrews Sports Medicine and Orthopaedic Center. Potential study participants then underwent a chart review of clinical and operative notes within our electronic health record. We determined potential study participants were eligible if (1) they underwent surgical reconstruction for an ACL tear plus any combination of injuries to the PCL, MCL, LCL, and/or structures of the PLC; (2) their MLKI was sustained during competitive participation in American football; and (3) they had a minimum follow-up of 2 years after surgery. Competitive participation in American football was defined as participation at the youth, club, junior high school, high school, collegiate, or professional level, where an athlete regularly participated in both practice and competition. Athletes were excluded if they participated at the recreational level with no organized competition. Additionally, we excluded potential participants if (1) they had absent or incomplete medical records, (2) their index procedure was a revision multiligament reconstruction, or (3) their index procedure was performed by a surgeon no longer associated with our institution because of concerns regarding consistency in longitudinal follow-up. Patients who had a history of ACLR before their MLKI and subsequent multiligament reconstruction were eligible for inclusion.

Patient and Injury Characteristics and Patient Evaluation

Chart-related data collected from the electronic health record included sex, age at surgery, injury/surgery laterality, body mass index (BMI), documented dislocation/subluxation, previous ipsilateral knee surgery history and

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type, whether the patient was a competitive American football athlete at the time of injury, level of competition and position, injury setting, and injury-to-surgery timing categorization as acute or chronic. Previous ipsilateral surgery history categories included none, ligamentous, meniscal/chondral, trauma, combination, or other. Injury-to-surgery timing categories were acute, if the index procedure at our institution occurred within 6 weeks from the date of injury, or chronic, if the operation occurred at or after 6 weeks from the time of the original injury.[15,27]

Operative data collected included ligaments involved, with the LCL and structures of the PLC being grouped together into 1 category (LCL/PLC). For each ligament involved, the operative technique and graft type were recorded, and ACLR number was recorded as primary, secondary (first revision), or tertiary (second revision). For patients with either or both the LCL and structures of the PLC involved, it was determined whether the peroneal nerve was injured based on preoperative physical examination findings of foot drop, weakness, and/or decreased dermatomal sensation. Last, index operations with other procedures performed not involving ligaments or structures of the PLC were categorized as meniscal, chondral, both, or other.

During chart review of potential participants as well as during follow-up, it was determined whether patients had subsequent ipsilateral knee surgeries at our institution or if they had any surgical procedures performed on their ipsilateral knee outside of our institution (see below). Subsequent ipsilateral knee operations were categorized as meniscal, chondral, arthritic/total knee arthroplasty, ligamentous, lysis of adhesions/manipulation under anesthesia, or hardware removal.

Surgical Management

All surgical procedures took place at 1 institution, a high-volume sports medicine practice (Andrews Sports Medicine and Orthopaedic Center), by 1 of 5 sports medicine fellowship-trained, board-certified orthopaedic surgeons (E.L.C., J.R.D., J.R.A., B.A.E., M.K.R.). Each surgeon had extensive past experience in treating complex sports injuries in competitive athletes, in addition to extensive experience in both single-ligament and multiligament knee reconstructions. Anatomically based reconstructions were performed to reconstruct torn ligaments. All ligamentous injuries as well as chondral and meniscal injuries were addressed during the same surgery. If the patient developed arthrofibrosis and had severely limited preoperative range of motion, preoperative physical therapy was typically pursued to increase motion before surgery. Although the surgeons performing these procedures had similar surgical techniques, small modifications in their techniques evolved over the long collection period for this study.

ACL Reconstruction. ACLRs were preferentially performed with a bone–patellar tendon–bone autograft using the Andrews technique, as previously described.[4,16] The graft was passed using a transtibial technique with femoral interference screw fixation followed by tibial screw fixation. Final graft tensioning was performed with the knee in full extension.

PCL Repair and Reconstruction. PCL injuries were treated via reconstruction, repair, or augmented repair. The augmented repair technique was described by Loeb et al.[30] PCL repair or repair with augmentation, in most cases, was performed for acute injury and either tibial or femoral end avulsion of the ligament, whereas reconstructions were typically done in more chronic cases or if the injured ligamentous substance was not sufficient for repair. For reconstructions, a single-bundle technique was used with a quadrupled hamstring tendon autograft in most cases. The gracilis and semitendinosus tendons were prepared in typical fashion and passed using a transtibial technique with suspensory femoral fixation. The final graft tensioning in the PCL reconstruction was typically performed with the knee in 90° of flexion.

MCL/Posteromedial Corner Repair. MCL injuries were addressed using direct/primary repair techniques similar to that described by Stannard et al.[44] Criteria for MCL repair were met in 4 general scenarios: (1) deep MCL avulsion from the proximal tibia (meniscotibial/coronary ligament injury) with synovial fluid extravasation into subcutaneous tissues; (2) proximal MCL tear accompanied by a capsular tear with the proximal ligament fragment flipped into the joint (Stener lesion); (3) proximal MCL injury that included the medial patellofemoral ligament with skeletonization of the medial distal femur; or (4) persistent valgus laxity after ACLR (± PCL reconstruction/repair/repair with augmentation) with capsular injury present. Repairs performed earlier in the study used expandable plastic anchors, whereas repairs performed more recently used soft/all-suture anchors.

LCL/PLC Repair/Reconstruction. LCL/PLC injuries were typically repaired when treated within 6 weeks of the injury. When reconstruction was required, the graft used varied based on which combination of ligaments was torn and was performed using techniques described by Clancy et al,[9] Dugas et al,[11] and LaPrade.[26] For an isolated PLC injury, the semitendinosus and gracilis were both used for reconstruction. For a PLC/LCL injury, the semitendinosus was used for the PLC with the gracilis used for the LCL. For the PLC/PCL injury, the quadrupled hamstring autograft was used for the PCL and a hamstring allograft used to reconstruct the PLC. The peroneal nerve was identified and released. Lateral meniscal injuries were typically repaired before PLC repair. The sequence of injuries addressed varied based on injuries present. The final tensioning sequence typically followed PCL, ACL, MCL, and finally PLC. Final tensioning of the PLC was done with the knee in 25° to 30° of flexion with valgus force.

Postoperative Rehabilitation

After surgical reconstruction, athletes were provided with a postoperative rehabilitation protocol. Although the goals of rehabilitation remained consistent throughout the study, the exact rehabilitative protocols and timing of each intervention evolved as best practices developed. Immediately postoperatively, patients used a drop lock brace set at 0° of extension or 10° of flexion (if the PLC and LCL were injured). All patients were permitted to be

<50% weightbearing using 2 crutches, with range of motion permitted between 0° and 45°. At approximately 12 days postoperatively, patients progressed to 75% weightbearing status. In the case of PLC/LCL injury, full extension was allowed at weeks 3 to 4. Between weeks 6 and 8 (depending on severity of injury), patients progressed to full weightbearing and were provided a functional ACL/PCL brace. Early- to midstage rehabilitation focused on restoring knee range of motion; regaining quadriceps, hamstring, and hip muscle strength; and improving neuromuscular control. Between approximately 6 and 8 months postoperatively, patients were permitted to gradually reenter sport-specific activity after achievement of a satisfactory clinical examination; satisfactory isokinetic testing, including demonstrating quadriceps strength >80% of the contralateral quadriceps; and a functional single-leg hop test >75% of the contralateral leg.

Patient-Reported Outcomes

Eligible patients (minimum postoperative time of 2 years) were contacted by telephone or by email to complete questionnaires to evaluate outcomes, including the IKDC form, an institution-specific surgical satisfaction survey, questions regarding subsequent ipsilateral knee surgeries, and an institution-specific RTS assessment. Informed consent was verbally obtained from each patient before completing the surveys. The IKDC form is a patient-reported questionnaire that includes 18 questions assessing pain, swelling, stiffness, instability, and activity limitations on a Likert scale, with higher scores indicating better knee function (score range, 0-100).[21] The IKDC form has been shown to be valid, reliable, and responsive to changes in knee function over time in individuals after knee injuries and surgeries, including ACLRs.[21,35] Our internally developed RTS assessment asks category-level questions to characterize each athlete's ability to RTS after surgery, the level of return, and contextual factors related to level of success in return. For the purposes of this study, we classified a successful RTS as an athlete returning to play in competitive American football at one's preinjury level or higher for any duration of time. Return to recreational play was not classified as successful RTS. The RTS assessment does not provide a numeric score. Similarly, the surgical satisfaction survey asks categorical questions related to satisfaction and is not numerically scored.

Statistical Analysis

We calculated summary statistics (means and standard deviations or counts and proportions) for baseline and/or follow-up demographic, clinical, surgical, and outcome data. We further compared baseline demographic, clinical, and surgical data between athletes with successful and unsuccessful follow-up using independent *t* test and chi-square test for continuous and categorical data, respectively. As a secondary analysis, we compared IKDC scores and the proportions successfully returning to preinjury sport between those who underwent 2-ligament surgical reconstruction and >2-ligament surgical reconstruction, as well as between those with ACL-MLKI injuries (no PCL involvement) and those with bicruciate MLKI injuries using independent *t* test (IKDC score) and chi-square test (RTS), respectively. Last, we examined the association between preinjury demographic variables (age and BMI) and IKDC score at follow-up or successful return to preinjury sport using linear (IKDC score) and logistic (RTS) regression. For all analyses, we considered P values <.05 to be statistically significant. All statistical analyses were performed using SPSS Version 28.0 (IBM Corp).

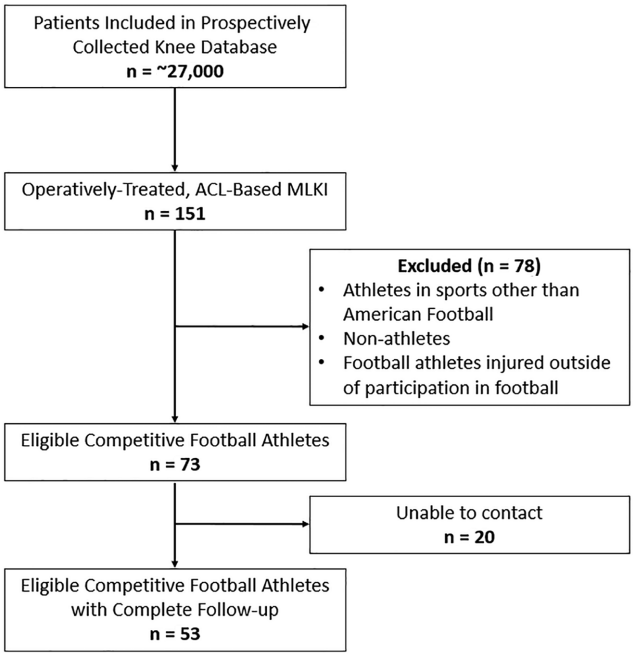

Figure 1. Flowchart of screened, excluded, and included participants. ACL, anterior cruciate ligament; MLKI, multiligament knee injury.

RESULTS

Cohort, Clinical, and Surgical Data

From our database of approximately 27,000 cases, 290 potential ACL-based MLKI reconstructions were identified. Of these 290 cases, 52 originally contained as physical charts were excluded because of missing chart information in the electronic health record. Following the application of all inclusion/exclusion criteria, 151 procedures were included, and 73 patients were identified as competitive football athletes eligible for this study (see Figure 1 for full inclusion and exclusion criteria). Follow-up was successfully completed with 53 of the football athletes (73%), with a mean follow-up time of 7.7 ± 4.0 years. Athletes with successful follow-up (n = 53) did not differ from athletes with unsuccessful follow-up (n = 20) in BMI (successful follow-up, 28.0 ± 5.6; unsuccessful follow-up, 27.9 ± 4.3; P = .48), the proportions of athletes with previous

TABLE 1
Patient and Injury Data[a]

Variable	Value
Age at surgery, y	
Mean ± SD	18.1 ± 2.7
Range	13.0-26.4
Sex	
Male	53 (100)
Female	0 (0)
Level of competition	
Junior high	3 (6)
High school	36 (68)
Collegiate	10 (19)
Professional	4 (8)
Laterality	
Left	30 (57)
Right	23 (43)
Previous ipsilateral knee surgery	
No	52 (98)
Yes	1 (2)
Ligamentous	1
Position	
Wide receiver	11 (21)
Running back	9 (17)
Offensive lineman	7 (13)
Defensive lineman	7 (13)
Linebacker	6 (11)
Free safety/defensive back	5 (9)
Other	8 (15)
Body mass index	
Mean ± SD	28.0 ± 5.6
Range	18.9-41.8
Surgery timing postinjury and category	
Mean ± SD, d	13.0 ± 12.7
Range, d	3-72
Acute, <6 wk	51 (96)
Chronic, ≥6 wk	2 (4)

[a]Data are presented as n (%) unless otherwise indicated. Percentages are based on included sample of n = 53.

ipsilateral knee surgery (successful follow-up, 2%; unsuccessful follow-up, 5%; P = .47), or the proportions of athletes with ≥ 3 ligaments surgically treated (successful follow-up, 26%; unsuccessful follow-up, 20%; P = .57). Those with successful follow-up were younger (age, 18.1 ± 2.7 years) than those with unsuccessful follow-up (age, 20.3 ± 3.9 years; P = .01).

Table 1 provides full descriptive and injury-related data. The mean age at operative treatment was 18.1 ± 2.7 years. Three athletes (6%) competed at the junior high level, 36 (68%) at the high school level, 10 (19%) at the collegiate level, and 4 (8%) at the professional level at the time of injury. A full breakdown of primary position played by each athlete is provided in Table 1. In total, 52 athletes (98%) had no operative treatment of their ipsilateral knee before their index procedure. The mean surgery timing between injury date and operative treatment was 13.0 ± 12.7 days, with 51 athletes (96%) undergoing surgery in the acute phase (<6 weeks after injury) (Table 1).

Table 2 provides full details regarding operative management and revision procedures. With respect to operative management, 39 athletes (74%) had 2 ligaments treated, 13 (25%) had 3 ligaments treated, and only 1 (2%) had all 4 ligaments treated. Combined ACL/MCL injuries were most common, with 22 athletes (42%) requiring operative treatment of these structures, followed by ACL and LCL/PLC injuries, present in 16 football athletes (30%). A complete breakdown of injuries is presented in Table 2. For ACLRs, the majority (n = 49; 92%) were performed using bone–patellar tendon–bone autografts. LCL/PLC injuries were addressed by repair in 22 of 23 lateral injuries. In these 23 athletes with lateral injuries, peroneal nerve deficits were evident in 6 athletes (26%), with 2 experiencing decreased sensation, 1 experiencing foot drop, and 3 demonstrating both sensory and motor deficits. In addition to ligament reconstruction, 29 athletes (55%) underwent additional procedures during reconstruction, the most common being isolated meniscal procedures in 25 athletes (86%), with lateral meniscal repair being the most common procedure (20/25 meniscal procedures). Additional details regarding reconstructive techniques and concomitant procedures are available in Table 2. After their index MLKI procedure, 9 patients (17%) underwent a subsequent knee surgery, while the majority (83%) did not. Of these 9 revision procedures, 4 (44%) were ligamentous revisions after failure of the primary ACL graft. Three of the 4 graft failures occurred during participation in sports. Other subsequent procedures are listed in Table 2. The mean time from initial multiligament reconstruction to revision procedure of any type was 9.4 months.

Patient-Reported Outcomes

Table 3 reports complete outcome-related data. At follow-up, the mean IKDC score of the cohort was 83.7 ± 15.5 (of 100) and the mean current patient-reported knee function (subitem of the IKDC form) was 7.7 ± 1.8 (of 10). In total, 50 athletes (94%) were either satisfied or very satisfied with their operation, and 49 (92%) thought their index procedure met or exceeded their expectations. Of the 53 athletes with follow-up data, 60% (32/53) successfully returned to preinjury participation in football. However, only 39 of the 53 attempted to return to participation in football, and of these, 32 (82%) were able to return to preinjury level of participation. Thus, only 7 athletes were unable to return specifically due to limitations from the MLKI surgery and recovery, including for specific reasons such as weakness, foot drop, instability, or psychological unpreparedness. Other reasons for unsuccessful RTS are shown in Table 3. For most athletes, return to practice occurred at a mean of 7.9 months postoperatively, and return to competition occurred at a mean of 9.5 months postoperatively. Of those who did successfully return (n = 32), 23 athletes (72%) indicated that their performance was about the same as or better than their performance before injury/surgery. Patients with peroneal nerve involvement at the time of their injury returned to play at a similar rate to the overall cohort, with 4 of 6 athletes (67%) successfully returning to sport.

TABLE 2
Surgical Data[a]

Variable	Value
No. of ligaments injured and treated	
2	39 (74)
3	13 (24)
4	1 (2)
Ligament combinations	
ACL + MCL	22 (42)
ACL + PCL	1 (2)
ACL + LCL/PLC	16 (30)
ACL + PCL + MCL	7 (13)
ACL + PCL + LCL/PLC	5 (9)
ACL + MCL + LCL/PLC	1 (2)
ACL + PCL + MCL + LCL/PLC	1 (2)
ACL (n = 53)	
Technique	
1°	52
2°	1
Graft source	
BPTB	49
Hamstring tendon	4
Graft type	
Autograft	53
PCL (n = 14)	
Technique	
Repair	2
Reconstruction	10
Repair with augmentation	2
Graft source (tendon)	
Hamstring	8
Semitendinosus	2
Tibialis anterior	1
Achilles	1
Graft type	
Autograft	10
Allograft	2
MCL (n = 31)	
Technique	
Repair	31
LCL/PLC (n = 23)	
Technique	
Reconstruction	1
Repair	22
Graft source	
Hamstring tendon	1
Graft type	
Autograft	1
Peroneal nerve involvement	6
Sensory deficit	2
Motor deficit	1
Both	6
Other procedures (n = 29)	
Meniscal	25
Meniscal/chondral	1
Other	3
Subsequent ipsilateral knee surgery	
No	44 (83)
Yes	9 (17)
Arthrofibrosis	1
Ligamentous	4
Meniscal	1
Meniscal/chondral	2
Hardware removal	1

[a]Data are presented as n (%) and all reported percentages are based on the included sample of n = 53. ACL, anterior cruciate ligament; LCL, lateral collateral ligament; MCL, medial collateral ligament; PCL, posterior cruciate ligament; PLC, posterolateral corner; BPTB, bone-patellar tendon-bone.

TABLE 3
Patient-Reported and Return-to-Sport Outcomes[a]

Variable	Value
Follow-up time, y	
Mean ± SD	7.7 ± 4.0
Range	2.0-17.1
IKDC score at follow-up	
Mean ± SD	84 ± 16
Range	15.5-24.1
Current knee function (0-10) at follow-up	
Mean ± SD	7.7 ± 1.8
Range	1.0-10.0
Satisfied or very satisfied	50 (94)
Met or exceeded expectations	49 (92)
Returned to sport, all reasons	
Yes	32 (60)
No	21 (40)
Reason for unsuccessful return to sport	
Limited by surgery	7/21
Graduated and not talented enough for the next level	11/21
Personal	3/21
Proportion of athletes able to return to preinjury sport who attempted to return after surgery	32/39

[a]Data are presented as n (%) unless otherwise indicated. IKDC, International Knee Documentation Committee Subjective Knee Form. Percentages are based on included sample of n = 53.

Subgroup Analyses and Predictors of Outcomes

Subgroup analysis revealed that there were no significant differences between athletes with 2 ligaments surgically treated versus those with >2 ligaments surgically treated in IKDC scores (2 ligaments, 83.7 ± 15.2; >2 ligaments, 83.9 ± 16.7; P = .96) or the proportions with successful RTS (2 ligaments, 62%; >2 ligaments, 57%; P = .77). Similarly, we found that there were no significant differences between athletes with ACL-MLKI injuries (no PCL involvement; n = 39) and those with bicruciate MLKI injuries (n = 14) in IKDC scores at follow-up (ACL only, 83.9 ± 15.3; bicruciate, 83.2 ± 16.6; P = .89) or the proportions with successful RTS (ACL only, 62%; bicruciate, 57%; P = .77). We further found no association between age at time of surgery or BMI at time of surgery and IKDC score at follow-up (age: model R^2 = 0.6%, P = .59; BMI: model R^2 = 4.1%, P = .15) or successful RTS (age, P = .12; BMI, P = .90).

DISCUSSION

MLKIs are complex injuries with the potential for devastating neurovascular consequences and a substantial negative effect on knee joint function, threatening an athlete's ability to return to competition after injury. In this study, we report longitudinal outcomes from a cohort of competitive American football athletes after MLKI reconstruction and demonstrate that most athletes are able to RTS after reconstruction and rehabilitation. We

found that this population of competitive athletes reported good patient-reported knee function, with a mean IKDC score of 84 ± 16, and high rates of satisfaction (94%) at follow-up, despite the severity of these injuries. Of athletes who attempted to return to football, 82% were able to RTS at the same level or higher. Additionally, there were no differences in knee-related function or RTS between athletes with 2 and >2 injured ligaments and no differences between those with ACL-MLKIs and bicruciate MLKIs. However, we found overall success in RTS and patient-reported function to lag behind reported RTS rates and function after isolated ACLR in athletes, emphasizing the severity of these injuries.[8,31,34,40]

Existing studies have reported a wide range of RTS success after surgical treatment of MLKIs.[12,18,37] Eranki and colleagues[12] reported that only 22% of competitive athletes returned to competition after MLKI, while, in contrast, Hirschmann and colleagues[18] reported that 78% of elite athletes were able to RTS after bicruciate MLKI reconstruction. However, these studies noted that only 33% were able to compete at their preinjury level of play.[12,18] Bakshi and colleagues[5] reported a similarly high RTS rate in NFL athletes after MLKI (64%; compared with 60% in our study) but found bicruciate tears to be associated with poorer RTS prognosis. In contrast, we found no differences in the proportions of successful RTS between ACL-only and bicruciate MLKIs in American football players. When considering all reasons for not returning to preinjury sport, we found that 60% of our football athletes were able to return at the same level of competition or better. However, when only considering those who attempted to return to preinjury level of participation in football, 82% were able to successfully return. Overall, however, RTS success after MLKI does appear to be worse than RTS success after isolated ACLR, with several studies reporting that ≥60% of competitive football athletes undergoing isolated ACLR successfully return to play.[8,31,34,40] Taken together, these findings suggest that there is further work to be done in optimizing MLKI surgical treatment and postoperative rehabilitation to maximize the potential for athletes to return to competition.

Our study also demonstrated high satisfaction (94%) and high patient-reported knee function scores, with no significant differences in outcomes based on previous knee surgeries, number of ligaments involved, or involvement of the PCL. Previous studies evaluating postoperative satisfaction after MLKI have reported that approximately 90% of athletes are satisfied with the outcomes of their surgery.[18,27] Similarly, other studies evaluating MLKI outcomes have shown no significant differences in postoperative function dependent on the combination of ligaments involved.[7,46] In previous systematic reviews and meta-analyses of patients who underwent surgical reconstruction in the acute period after MLKI, the mean IKDC scores were 68 and 69,[10,32] compared with our mean score of 84, demonstrating that the athletes in our study may demonstrate better knee-related function compared with the average patient after MLKI. The functional and high-level athletic status before surgery of the athletes included in our study may have played a part in the high function scores postoperatively. Additionally, the relatively young age of our cohort (18.1 ± 2.7 years) when compared with that of other cohorts (34.5 ± 13.6 years; range, 19-47 years) in similar studies may also have affected our outcomes.[10,27] Younger age at surgery has been previously associated with higher patient-reported outcome scores and increased likelihood of successful RTS after isolated ACLR, and similar trends may be seen after multiligament reconstruction.[2,47] Other large studies evaluating outcomes after MLKI have utilized the Lysholm knee scoring scale, Western Ontario and McMaster Universities Osteoarthritis Index scores, and Tegner activity score, which are not directly comparable with the IKDC score.[13,27] IKDC scores from athletes with long-term follow-up after isolated ACLR have been reported as high as 97 ± 10.[6] These findings further demonstrate the injury severity of an MLKI, as well as the need for further work to maximize postoperative function.

Our study also demonstrated rates of postoperative complications and revisions comparable with those previously reported in other studies. In our study, 17% of athletes required subsequent surgery, and revision rates after MLKI in active populations have been reported to range from 14% to 20%.[18,27,45] Common postoperative complications requiring revision described in the literature include ligament failure and arthrofibrosis.[10,27] We observed a subsequent ligament graft failure rate of 8%, compared with 5% described in other similar studies evaluating outcomes in competitive athletes.[27] Previous studies have described outcomes in recreational athletes, and we argue that our rate of ligament failure is comparable, given that many of our athletes returned to play football at a competitive level, a sport where ligamentous tears, such as ACL tears, are frequent in athletes with no previous history of MLKI. Arthrofibrosis was observed in 2% of athletes in this study. Historically, surgical intervention in the acute period after injury has been associated with development of arthrofibrosis[14,37,42,43]; however, almost all athletes in our study were treated in the acute phase, and only 1 athlete subsequently developed arthrofibrosis. Rates of arthrofibrosis after acute reconstruction have been reported at 9% in previous studies.[27,37] We believe aggressive postoperative rehabilitation may contribute to the low rates of postoperative arthrofibrosis seen in our competitive athletes.

Several factors may have contributed to the outcomes that we demonstrated in this study, including level of surgical training and experience, operative techniques used, and rehabilitation expertise. First, our study involved procedures performed by fellowship-trained surgeons with high volumes of both multiligament reconstructions and single-ligament reconstructions in competitive athletes, which allows for the development of both efficiency and expertise. Surgeries performed by surgeons who perform high volumes of specific procedures have been linked to more accurate graft placement during knee reconstructions, as well as decreased complications and revision rates when compared with surgeons performing lower case volumes.[19,28,48] Fellowship training has been shown to dramatically increase case volume in sports injuries.[20] However, MLKIs are an area many sports medicine fellowship-trained surgeons feel less confident in

treating.[49] Ensuring MLKIs are treated by a surgeon trained in a fellowship program that provides adequate exposure to complex cases, including MLKIs, is important. Additionally, each patient was treated in a single-stage reconstruction procedure with or without concomitant repairs. The combination of experience and single-stage operations allowed for decreased total operative and tourniquet time, potentially reducing the risk of soft tissue damage and further complications. Longer total operative and tourniquet times have been associated with increased complication rates in multiple orthopaedic procedures, including multiligament knee reconstructions.[3,39] Last, patients are set up to have optimal recovery and rehabilitation in the immediate postoperative period. Our institution has worked closely with physical therapists who also have significant experience in complex knee injuries to optimize postoperative rehabilitation, creating a balance of early motion with protection of the reconstruction. Furthermore, our team of physical therapists has extensive experience in the rehabilitation of high-level competitive athletes, critical for ensuring successful RTS.

Limitations

There are several limitations that should be recognized related to our study. First, our study lacks baseline/presurgical data, preventing us from evaluating the effect of preoperative function on outcomes and determining if clinically relevant improvements in knee-related function from before to after surgery are present. Second, the collection of outcome data in this cohort may be subject to biases. Certain data points within our assessment of RTS, such as factors influencing a player's decision to pursue a return to athletics or a player's perception of one's preoperative and postoperative performance, may be subject to recall bias. Follow-up time for each athlete varied, thus requiring each athlete to accurately remember details about RTS, which may have occurred many years previously. Additionally, our assessment of surgical satisfaction is affected by confirmation bias, as patients reported their satisfaction scores to researchers associated with the clinic. However, recall bias is unavoidable in a retrospective study, and our instruments are designed to mitigate bias and allow for the most accurate reporting possible. Third, our study lacks radiographic data at follow-up. Although collecting radiographic data was beyond the scope of this study, we were unable to evaluate outcomes related to mid- to long-term joint health after surgical treatment of MLKIs in these athletes. Longer follow-up (beyond the mean follow-up time of 7.7 years in the current study), along with the collection of radiographic data, is needed to better understand long-term knee function and health in these young athletes. Last, our study includes a relatively small sample, when considered in comparison with outcome studies in other sports medicine and orthopaedic injury patient populations. In this study, there were 52 patients who underwent operative treatment for an MLKI but had incomplete clinical documentation, limiting the sample size eligible for follow-up in our study. However, given the relative rarity of MLKIs, to our knowledge, the current study represents the largest MLKI outcome study in a cohort of competitive American football athletes. Despite these limitations, our study provides clinically relevant information regarding prognosis after MLKI for American football athletes. Future studies should aim to further optimize surgical management and rehabilitation after an MLKI, as well as better identify potential factors that increase an athlete's risk of complications, poor knee-related function, failure to RTS, or poor long-term knee joint health.

CONCLUSION

This study demonstrated high levels of satisfaction and good patient-reported knee function after operative treatment of MLKIs in competitive American football athletes, supporting treatment of these injuries via acute, single-stage reconstruction by a fellowship-trained surgeon followed by optimized postoperative rehabilitation. Despite the severity of these injuries, the rate of successful RTS for this cohort was high but still lower in comparison with rates historically reported for isolated ACLR. This study further demonstrated that there were no significant differences in outcomes in athletes with bicruciate MLKI injuries and athletes with >2 injured ligaments when compared with those with an ACL-only MLKI and 2 ligaments injured, respectively. Future studies are needed to identify predictors of success and better understand optimal treatments for each injury pattern.

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